

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Article Review

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 4: Students will be able to analyze and evaluate advanced deep learning techniques and architectures, interpret research findings, and apply appropriate models to real-world problems. They will be able to critically review research articles, compare methodologies, and justify suitable deep learning solutions. (BL-6)

Assessment Tool Weightage: 30%.

QUESTIONS:

Q1. Article Identification and Summary

Select a recent peer-reviewed research article related to Deep Learning, such as CNNs, Transfer Learning, ResNet, DenseNet, RNNs, LSTM, Autoencoders, or other advanced architectures. Provide the article title, authors, publication venue, year, and research objective. Summarize the main problem addressed, proposed approach, and key findings in your own words.

Q2. Problem Statement and Motivation

Explain the real-world problem addressed by the selected article and analyze why the problem requires a deep learning solution. Identify the limitations of existing approaches discussed by the authors and explain the motivation for the proposed method.

Q3. Methodology and Architecture Review

Critically review the methodology used in the article. Describe the dataset, preprocessing, model architecture, important layers or components, training strategy, and evaluation metrics. Illustrate the proposed architecture or workflow and explain the role of its major components.

Q4. Results, Comparison and Critical Analysis

Analyze the experimental results reported in the article. Compare the proposed method with the baseline or existing methods using the reported performance measures. Discuss the strengths, limitations, computational considerations, and validity of the results. Identify at least two possible improvements or future research directions.

Q5. Application and Personal Evaluation

Explain how the reviewed deep learning method can be applied to a practical industry problem. Propose a suitable implementation or extension and justify your choices. Conclude the review with your overall evaluation of the article, including its contribution, practical relevance, and scope for future work.

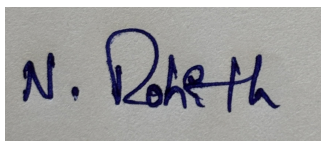
Article Review Submission Guidelines

- Choose one recent and relevant peer-reviewed research article in Deep Learning.
- Attach or provide the complete citation/reference of the selected article.
- The review must be written in the student's own words and should demonstrate critical analysis.
- Include architecture diagrams, tables, graphs, or screenshots where they improve explanation.
- Use appropriate academic referencing and avoid plagiarism.
- Recommended review length: 4–6 pages, excluding the reference page.

Code of Conduct:

I **NALIPI REDDY ROHITH/192425115**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Understanding & Summary	20%	Comprehensive and accurate understanding of the article, problem, objectives, and contributions.	Good understanding with minor omissions or inaccuracies.	Basic understanding; important details are missing.	Poor understanding or significant misinterpretation.
Critical Review of Methodology	25%	Thoroughly evaluates methodology, architecture, datasets,	Appropriate analysis with minor gaps.	Limited analysis with several unexplained aspects.	Little or no meaningful analysis of methodology.

		training, and evaluation choices.			
Analysis of Results	20%	Accurately interprets results, comparisons, metrics, strengths, and limitations.	Good interpretation with reasonable comparison.	Basic interpretation with limited comparison.	Incorrect, unsupported, or missing interpretation.
Application & Proposed Improvements	20%	Practical, innovative application with well-justified improvements and future directions.	Relevant application and feasible improvements.	Basic application with limited justification.	Unclear, impractical, or unsupported proposal.
Presentation & Referencing	15%	Well-organized, professional presentation with clear diagrams and correct academic referencing.	Clear presentation with minor formatting or referencing issues.	Adequate presentation with inconsistent formatting/referencing.	Poor organization, unclear presentation, or inadequate referencing.

Suggested Article Review Structure

1. Article Citation
2. Abstract / Executive Summary
3. Research Problem and Motivation
4. Methodology and Deep Learning Architecture
5. Dataset and Experimental Setup
6. Results and Comparative Analysis
7. Strengths and Limitations
8. Proposed Improvements / Future Scope
9. Practical Industry Application
10. Conclusion and References

Article Review

Q1. Article Identification and Summary

1. Article Citation

Title: *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*

Authors: Rukhsar Sabir and Tahir Mehmood

Journal: *Scientific Reports*

Volume: 14

Article Number: 29704

Year: 2024

DOI: 10.1038/s41598-024-75143-4

The article studies deep-learning-based classification of melanoma using different neural-network architectures.

2. Abstract / Executive Summary

The article focuses on automatic melanoma classification from skin images. The researchers compare three neural-network approaches: a basic CNN, ResNet-18 and EfficientNet-B0.

The main objective is to determine which architecture can provide accurate classification of benign and malignant skin lesions. The study reports that EfficientNet-B0 gives the strongest overall performance.

3. Research Problem and Motivation

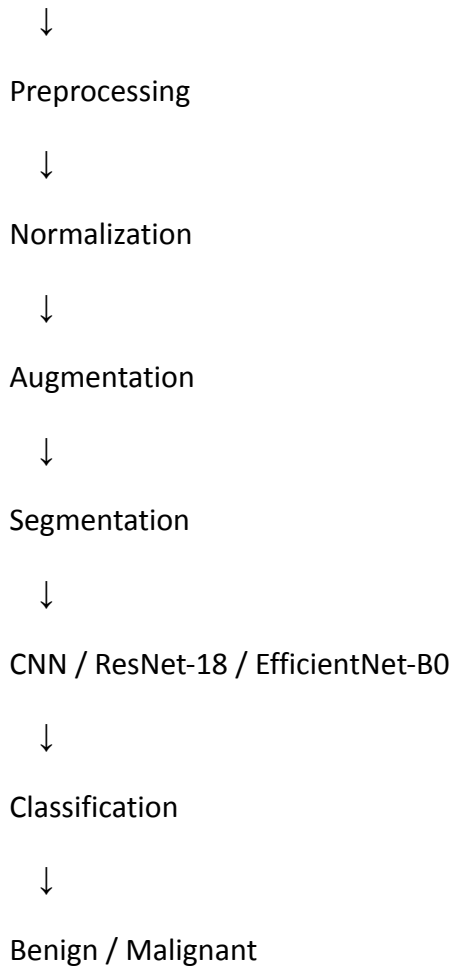
Melanoma is a serious type of skin cancer. Early identification can improve treatment outcomes. However, visual diagnosis can be difficult because different skin lesions can have similar appearances.

The authors are motivated to develop an automated image-analysis system that can assist in melanoma detection.

4. Methodology and Deep Learning Architecture

The general workflow is:

Skin Image



The CNN extracts visual features using convolutional layers. ResNet-18 uses residual connections, while EfficientNet-B0 uses efficient convolutional blocks.

5. Dataset and Experimental Setup

The researchers use a melanoma image dataset containing thousands of skin images. Images are processed and resized before being provided to the models.

The models are trained and evaluated using training and testing data. Performance is measured using accuracy, sensitivity, specificity, precision and F1-score.

6. Results and Comparative Analysis

The reported results show that EfficientNet-B0 performs better than the other evaluated architectures.

Model	Accuracy
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CNN	80%
ResNet-18	87%
EfficientNet-B0	97%

EfficientNet-B0 also achieved very high sensitivity, which is important for detecting malignant cases.

7. Strengths and Limitations

Strengths:

- Compares multiple architectures.
- Uses several performance metrics.
- Addresses an important healthcare problem.
- EfficientNet-B0 provides strong results.

Limitations:

- Dataset diversity may affect generalization.
- Real clinical images can differ from research datasets.
- Additional external validation is required.

8. Proposed Improvements / Future Scope

Future work could include:

1. Testing on larger datasets from multiple hospitals.
2. Using Explainable AI such as Grad-CAM.
3. Including more diverse skin types.
4. Optimizing the model for mobile or real-time applications.

9. Practical Industry Application

The model can be developed as a computer-aided skin cancer screening system.

Dermoscopic Image



Preprocessing



EfficientNet-B0



Prediction



Benign / Malignant



Doctor Review

It can help dermatologists prioritize suspicious cases.

10. Conclusion and References

The article demonstrates that advanced CNN architectures can achieve strong melanoma-classification performance. EfficientNet-B0 produced the best reported results.

Reference:

Sabir, R., & Mehmood, T. (2024). *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*. Scientific Reports, 14, 29704.

Q2. Problem Statement and Motivation

1. Article Citation

Title: *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*

Authors: Rukhsar Sabir and Tahir Mehmood

Journal: *Scientific Reports*

Year: 2024.

2. Abstract / Executive Summary

The research investigates the use of deep learning for melanoma detection. CNN, ResNet-18 and EfficientNet-B0 are compared to identify an effective architecture for skin-lesion classification.

The research shows that deeper and more efficient architectures can provide better classification than a basic CNN.

3. Research Problem and Motivation

The major problem is the difficulty of distinguishing melanoma from benign lesions using visual information alone.

Traditional methods often depend on manually designed image features. Deep learning can automatically learn useful patterns from the images.

4. Methodology and Deep Learning Architecture

The study uses multiple CNN-based models.

Input Image



Image Processing



Feature Extraction



CNN Architecture



Classification

The comparison allows the researchers to analyze how architecture affects classification performance.

5. Dataset and Experimental Setup

The dataset consists of melanoma-related images. The images are normalized and processed before training.

The models are trained using the prepared dataset and evaluated using multiple classification metrics.

6. Results and Comparative Analysis

The reported accuracy comparison is:

Architecture	Accuracy
CNN	80%
ResNet-18	87%
EfficientNet-B0	97%

The results indicate that EfficientNet-B0 provides the best overall classification.

7. Strengths and Limitations

Strengths:

- Systematic model comparison.
- High performance from EfficientNet-B0.
- Multiple evaluation metrics.

Limitations:

- Results depend on the selected dataset.
- Generalization to other clinical environments requires further testing.
- Computational requirements may be significant.

8. Proposed Improvements / Future Scope

The research can be improved by:

- Using larger multi-hospital datasets.
- Applying transfer learning.
- Using attention mechanisms.
- Adding Explainable AI.
- Performing external clinical validation.

9. Practical Industry Application

Hospitals and dermatology clinics could use the system as a **decision-support application**.

A dermatologist could upload an image and receive a predicted risk category.

The final diagnosis should remain with a qualified medical professional.

10. Conclusion and References

The research addresses an important medical problem and demonstrates the usefulness of deep learning for melanoma classification.

Reference:

Sabir, R., & Mehmood, T. (2024). *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*. Scientific Reports, 14, 29704.

Q3. Methodology and Architecture Review

1. Article Citation

Sabir, R., & Mehmood, T. (2024). *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*. **Scientific Reports**, 14, 29704.

2. Abstract / Executive Summary

The article evaluates different CNN architectures for melanoma classification. The researchers investigate how CNN, ResNet-18 and EfficientNet-B0 perform on skin-cancer images.

3. Research Problem and Motivation

The objective is to automatically identify malignant skin lesions from images. Accurate classification is important because early melanoma detection can improve medical outcomes.

4. Methodology and Deep Learning Architecture

The major architectures are:

CNN

Uses convolution and pooling operations to extract image features.

ResNet-18

Uses residual/skip connections:

Input



Convolution



Convolution



+ ← Skip Connection



Output

EfficientNet-B0

Uses efficient convolutional blocks and compound scaling to achieve a balance between performance and computational requirements.

5. Dataset and Experimental Setup

The research uses a melanoma skin-cancer image dataset. Images are preprocessed and standardized before being given to the neural networks.

The models are evaluated using metrics such as:

- Accuracy
- Precision
- Sensitivity
- Specificity
- F1-score
- MCC

6. Results and Comparative Analysis

EfficientNet-B0 achieved the best reported accuracy.

Model	Accuracy	Sensitivity	Specificity
CNN	80%	64%	95%
ResNet-18	87%	82%	93%
EfficientNet-B0	97%	99%	93%

7. Strengths and Limitations

Strengths:

- Multiple architectures are evaluated.
- EfficientNet achieves excellent performance.
- Multiple metrics are used.

Limitations:

- The study depends on one primary dataset.
- Clinical deployment requires further validation.
- Deep learning models can require substantial computational resources.

8. Proposed Improvements / Future Scope

Possible improvements include:

- Larger datasets.
- Transfer learning.
- Attention-based models.
- Explainable AI.
- External validation.
- Real-time deployment.

9. Practical Industry Application

The architecture can be integrated into a medical image analysis platform.

Image Upload



Preprocessing



EfficientNet



Feature Extraction



Classification



Prediction

The prediction can support doctors during screening.

10. Conclusion and References

The methodology demonstrates that EfficientNet-B0 can provide better melanoma classification performance than the compared CNN and ResNet-18 models.

Reference: Sabir, R., & Mehmood, T. (2024). *Scientific Reports*, 14, 29704.

Q4. Results, Comparison and Critical Analysis

1. Article Citation

Sabir, R., & Mehmood, T. (2024). *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*. *Scientific Reports*, 14, 29704.

2. Abstract / Executive Summary

The article compares CNN, ResNet-18 and EfficientNet-B0 for melanoma classification and analyzes their performance using several evaluation measures.

3. Research Problem and Motivation

The study aims to improve automated detection of melanoma. A model with high sensitivity is particularly useful because failing to identify malignant lesions can be dangerous.

4. Methodology and Deep Learning Architecture

The models receive processed skin images and learn visual features.

The workflow is:

Images



Preprocessing



Feature Extraction



CNN / ResNet / EfficientNet



Classification

5. Dataset and Experimental Setup

The study uses a melanoma image dataset and performs preprocessing before training. Multiple metrics are used to provide a more complete assessment than accuracy alone.

6. Results and Comparative Analysis

Model	Accuracy	Precision	F1-score
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CNN	80%	93%	76%
ResNet-18	87%	91%	86%
EfficientNet-B0	97%	95%	97%

EfficientNet-B0 shows the strongest overall performance.

Its high sensitivity is particularly important for medical screening.

7. Strengths and Limitations

Strengths

- High classification accuracy.
- Comparison with baseline models.
- Multiple evaluation metrics.
- Important real-world healthcare application.

Limitations

- Dataset bias can affect results.
- The model may not perform equally well on unseen populations.
- Computational requirements should be considered.
- Clinical trials and external validation are necessary.

8. Proposed Improvements / Future Scope

Two important improvements are:

Improvement 1: Use multiple independent medical datasets to test generalization.

Improvement 2: Add Grad-CAM or another Explainable AI technique so doctors can understand the regions responsible for the prediction.

Other improvements include model compression, larger datasets and real-time deployment.

9. Practical Industry Application

The model can be incorporated into a hospital's dermatology decision-support system.

Patient Image



AI Model



Risk Score



Doctor



Final Diagnosis

This can help reduce screening time and prioritize suspicious cases.

10. Conclusion and References

The results indicate that EfficientNet-B0 significantly outperforms the basic CNN and ResNet-18 on the reported dataset. However, further external clinical validation is needed before real-world medical deployment.

Reference:

Sabir, R., & Mehmood, T. (2024). *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*. Scientific Reports, 14, 29704.

Q5. Application and Personal Evaluation

1. Article Citation

Title: *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*

Authors: Rukhsar Sabir and Tahir Mehmood

Venue: Scientific Reports

Year: 2024

Article: 29704

2. Abstract / Executive Summary

The research demonstrates how different deep learning architectures can classify melanoma skin-cancer images. EfficientNet-B0 provides the strongest reported performance among the evaluated models.

3. Research Problem and Motivation

The motivation is to provide an automated system that can help identify suspicious skin lesions. Such systems can support doctors and potentially improve early screening.

4. Methodology and Deep Learning Architecture

A practical implementation could use:

Dermoscopic Image



Resize + Normalize



Data Augmentation



EfficientNet-B0



Global Feature Representation



Classification Layer



Benign / Malignant

The EfficientNet model can be initialized with pre-trained weights and fine-tuned on medical images.

5. Dataset and Experimental Setup

For an industry implementation, the system should be trained using a large, diverse dataset collected from different hospitals and imaging devices.

The data should be divided into:

- Training set
- Validation set
- Independent test set

Performance should be measured using accuracy, sensitivity, specificity, precision and F1-score.

6. Results and Comparative Analysis

The article reports:

Model	Accuracy
CNN	80%
ResNet-18	87%
EfficientNet-B0	97%

Therefore, EfficientNet-B0 would be a strong candidate for an industry prototype.

7. Strengths and Limitations

Strengths

- Strong performance.
- Efficient architecture.
- Suitable for image classification.
- Can be adapted using transfer learning.

Limitations

- Research-dataset performance may not equal clinical performance.
- Different cameras and imaging conditions can affect predictions.
- Medical AI requires strong validation and monitoring.
- Incorrect predictions could have serious consequences.

8. Proposed Improvements / Future Scope

I would extend the system with:

1. **Grad-CAM:** Display the region that influenced the prediction.
2. **Multi-hospital dataset:** Improve generalization.
3. **Mobile deployment:** Provide preliminary screening through a mobile or web application.
4. **Continuous monitoring:** Track model performance after deployment.
5. **Human-in-the-loop:** Require a dermatologist to make the final decision.

9. Practical Industry Application

A healthcare company could develop a web application:

Doctor uploads image



Image preprocessing



EfficientNet-B0



Melanoma probability



Heatmap explanation



Doctor reviews result

This could help dermatologists prioritize high-risk cases while keeping the final decision with the medical professional.

10. Conclusion and References

Personal Evaluation

In my opinion, the article is valuable because it demonstrates how modern CNN architectures can improve automated melanoma classification. The comparison between CNN, ResNet-18 and EfficientNet-B0 makes the study useful for understanding the effect of architecture on medical image classification.

The reported **97% accuracy and 99% sensitivity of EfficientNet-B0** demonstrate strong potential.

However, the model should undergo extensive external validation and explainability testing before clinical deployment.

Final Conclusion

The article makes a useful contribution to medical image analysis by demonstrating the potential of efficient deep learning architectures for melanoma detection. Its practical scope can be expanded through larger datasets, Explainable AI, external validation and real-world clinical integration.

Reference

Sabir, R., & Mehmood, T. (2024). *Classification of melanoma skin Cancer based on Image Data Set using different neural networks*. **Scientific Reports**, **14**, 29704.